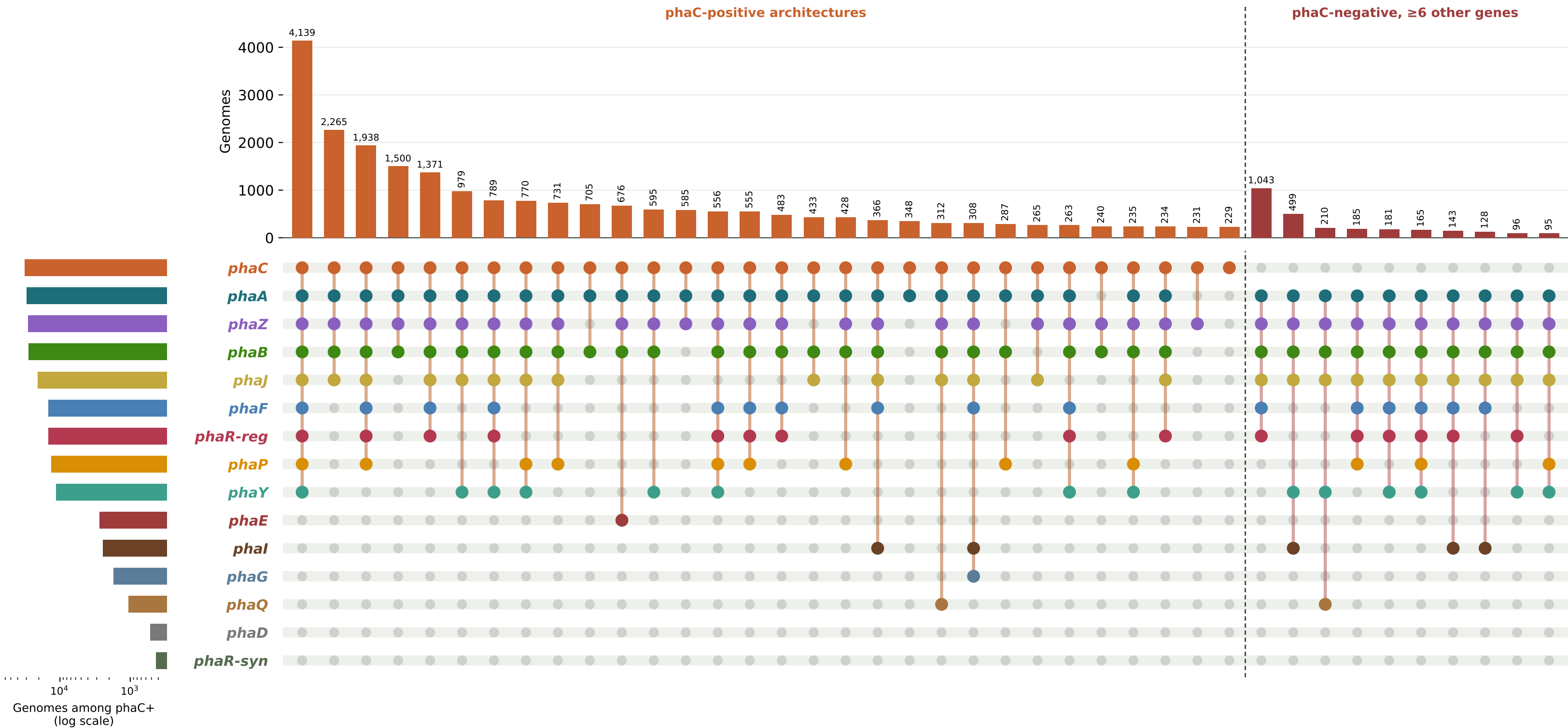


PHA Pathway Gene Architectures: Presence, Co-occurrence, and PhaC-Negative Outliers

Left: top 30 architectures among 31,464 phaC-positive genomes (580 distinct architectures total) | Right: top 10 of 196 architectures with phaC absent but >5 other PHA genes present



Architecture = the set of PHA pathway genes co-occurring in one genome (pipeline/pathway_architecture.py). Row order and left-panel totals are scoped to phaC-positive genomes only. Dot color = gene identity (matches row label and left bar); hollow/faint dot = absent. Right-hand columns (red) carry a substantial accessory-gene suite but no phaC hit -- a genuinely divergent/incomplete pathway, or a search miss. Excludes mislabeled references across all 13 families (worst: phaE/phaG/phaR-syn -- see FAMILY_BAD_QUERIES in pipeline/pathway_architecture.py).